FLAGSHIP 01 · BIONIC EXOSKELETON SERIES

LC Power-U/A (Spec: LC Power-U/P)

Passive Upper Limb Exoskeleton · Overhead Bionic Armor

Overhead Bionic Armor. Pure Mechanical Relief.

Engineered with a pure mechanical bionic structure for precise directional assistance, the LC Power-U/A is lightweight, highly reliable, and adaptable to complex industrial environments. Featuring multi-level adjustments to fit diverse body types, it is purpose-built for automotive assembly, power maintenance, construction, aircraft repair, and agricultural picking - dramatically reducing shoulder and arm muscle load while elevating operational efficiency and all-day comfort.

01 Featherlight Carbon Frame

Aerospace carbon fiber paired with wear-resistant Oxford cloth delivers unmatched rigidity at sub-2kg weight, ensuring unrestricted arm mobility.

02 Intrinsically Safe Zero-Power

100% mechanical elastic energy storage eliminates battery maintenance, charging downtime, and spark hazards across continuous multi-shift operations.

03 High-Efficiency Musculoskeletal Offloading

Slashing deltoid muscle strain by up to 8kg, directly extending worker endurance and preventing chronic shoulder disorders.

04 Minimalist Industrial Durability

High-strength anodized aluminum joints withstand rigorous impacts, dust, and continuous industrial wear.

LC Power-U/A · OFFICIAL HARDWARE VIEW



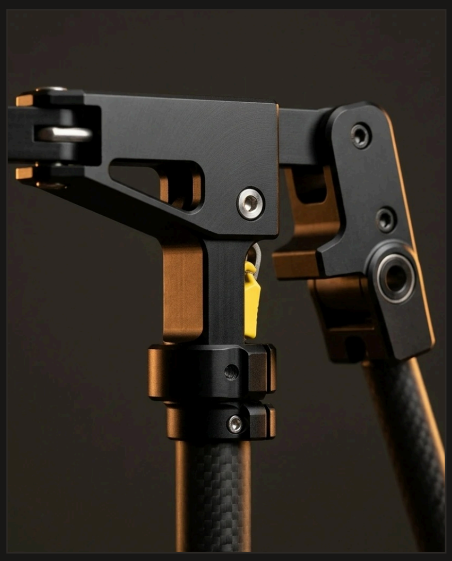
KEY METRIC 01
8kg
Maximum Overhead Assist Capacity

KEY METRIC 02
< 2kg
Ultra-Light Aerospace Carbon Fiber Spine

KEY METRIC 03
100%
Passive Zero-Power Operation (Zero Battery Risk)

KEY METRIC 04
4 Angles
Multi-Gear Articulation (50°, 70°, 90°, 110°/115°)

MATERIAL SCIENCE & MICRO-ARCHITECTURE · STRUCTURAL CRAFTSMANSHIP DETAILS



AVIATION ALUMINUM ALLOY · 4-ANGLE LOCK

4-Angle Semi-Automatic Mechanical Shoulder Joint

CNC-machined aviation aluminum alloy housing encloses a 3-gear, 4-angle (50°, 70°, 90°, 110°/115°) semi-automatic locking ratchet that engages silently as the operator raises their arms overhead and disengages smoothly on descent.



T700 AEROSPACE CARBON FIBER

Aerospace Carbon Fiber Telescoping Spine

Multi-directional woven aerospace carbon fiber spine resists torsional flex while transferring up to 8kg of overhead tool weight down to the pelvic belt.



OXFORD OUTER + 3D HONEYCOMB MESH

Quick-Release Armrest Cuffs & 3D Air-Mesh

Engineered with tear-resistant industrial Oxford outer fabric and 3D honeycomb air-mesh inner lining for rapid emergency release and 8-hour thermal ventilation.

ANATOMICAL & MECHANICAL SUBSYSTEMS (5 NODES)

1. **4-Gear Shoulder Joint [Glenohumeral Articulation Unit]**
   Semi-automatic mechanical assist engages instantly as arms elevate, with locks at 50°, 70°, 90°, 110°/115°.
2. **Quick-Release Chest & Dorsal Harness [Thoracic Cross-Lock Harness]**
   Ergonomic snap buckles enable rapid donning and doffing in under 15 seconds.
3. **Telescoping Carbon Back Bar [Dorsal Carbon Spine]**
   High-tensile carbon fiber back spine seamlessly adjusts to different torso lengths.
4. **Quick-Release Armrest Cuffs [Humeral Support Cradle]**
   Cuffed ergonomic armrests with yellow buckle and blue quick-release tab provide immediate escape.
5. **Contoured 3D Lumbar Belt [Pelvic Load-Transfer Girdle]**
   Breathable air-mesh waist belt distributes harness load evenly across the pelvic girdle.

FIELD DEPLOYMENT | AUTOMOTIVE FINAL ASSEMBLY LINE | OVERHEAD CHASSIS STATION



Underbody Automotive Assembly & Overhead Torque Operations

Inside modern automotive assembly plants, technicians execute thousands of overhead fastening cycles per shift beneath elevated vehicle chassis. The LC Power-U/A locks in mechanical support at 50° to 115°, neutralizing the weight of pneumatic torque wrenches and eliminating deltoid fatigue without requiring a single watt of electrical power.

Up to 8kg Offload	< 15 Seconds	0 Hours (100% Passive)
Deltoid EMG Reduction	Donning / Doffing Time	Power & Charging Downtime

COMPLETE FACTORY TECHNICAL SPECIFICATIONS · LC Power-U/A (100% UNABBREVIATED)

ENGINEERING GROUP	PARAMETER	VERIFIED FACTORY SPECIFICATION
IDENTITY & COMPLIANCE	**Model**	LC Power-U/A (Spec Code: LC Power-U/P)
IDENTITY & COMPLIANCE	**Color**	Industrial Matte Black
IDENTITY & COMPLIANCE	**Target Users**	Workers with prolonged arm-raising and overhead tasks
IDENTITY & COMPLIANCE	**Applications**	Automotive assembly, power maintenance, construction, aircraft repair, agricultural harvesting
MATERIALS & ERGONOMICS	**Structural Materials**	Carbon fiber + aviation aluminum alloy
PERFORMANCE & MECHANICS	**Assistance Capacity**	Max. 8kg
PERFORMANCE & MECHANICS	**Locking Mode**	Semi-automatic mechanical locking
PERFORMANCE & MECHANICS	**Assistance Gears**	3 gears (4-angle engagement)
PERFORMANCE & MECHANICS	**Assistance / Unlocking Angles**	50°, 70°, 90°, 110°/115°
MATERIALS & ERGONOMICS	**Outer Fabric**	Wear-resistant Oxford fabric
MATERIALS & ERGONOMICS	**Lining Fabric**	Breathable 3D mesh fabric
DIMENSIONS & WARRANTY	**Sizes**	Standard Size / Plus Size
DIMENSIONS & WARRANTY	**Net Weight**	< 2kg
DIMENSIONS & WARRANTY	**Package Dimensions**	50 x 40 x 12cm
DIMENSIONS & WARRANTY	**Optional Accessories**	Ergonomic neck support collar
DIMENSIONS & WARRANTY	**Warranty**	1-Year Official Industrial Warranty

DIRECT FACTORY PROCUREMENT & PILOT DEPLOYMENT - LC Power-U/A

Manufacturer: Guangdong Langcheng Technology Co., Ltd. | R&D Base: Intelligent Protection Technology R&D Center (GDUT Institute of CNC Equipment Collaborative Innovation)

Hotline / WhatsApp: +86 159-8917-5854 | Engineering Email: 1094099100@qq.com | Warranty: 1-Year Official Industrial Manufacturer Warranty Across All Hardware